Code No: 113AU **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year I Semester Examinations, September/October - 2023
ELECTRONIC DEVICES AND CIRCUITS
(Common to EEE, ECE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a) Write the applications of photo diode. [2]
b) Write the effect of temperature on diode characteristics. [3]
c) Define Transformer utilization factor [2]
d) What is the need of a capacitive filter? [3]
e) Which of the BJT configurations are suitable for impedance matching applications and why? [2]
f) Give the other name of early effect and explain the same [3]
g) Mention the need for biasing? [2]
h) What is emitter follower? Draw the circuit diagram of CC configuration. [3]
i) Why FET is called voltage controlled device? [2]
j) What do you mean by pinch-off in JFET? [3]

PART – B

(50 Marks)

2.a) Explain V-I characteristics of Zener diode.
b) A Ge diode has a saturation current of 10μA at 300° K. Determine the saturation current at 400°K. [5+5]

OR

3.a) With neat diagram, explain the construction and working of Varactor Diode.
b) Explain Transition and Diffusion Capacitances of a PN Junction diode. [5+5]

4.a) Explain Zener diode shunt regulator circuit.
b) Derive the expression for Ripple factor for Full Wave Rectifier with L-Section filter. [5+5]

OR

5.a) A full-wave single phase rectifier employs a pi- section filter consisting of two 4 μF capacitances and a 20 H choke. The transformer voltage to the center tap is 300 V rms. The load current is 500 mA. Calculate the dc output voltage and the ripple voltage. The resistance of the choke is 200Ω.
b) Show that the maximum efficiency of half wave rectifier is 40.6% [5+5]

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6.a) A transistor is operated at a forward current of 2μA and with the collector open circuited. Calculate the junction voltages V_C and V_E, the collector to emitter voltage V_{CE} assuming I_{CO} = 2μA, I_{EO} = 1.6μA and αN = 0.98.

b) Analyze the working mechanism of CB configuration of BJT [5+5]

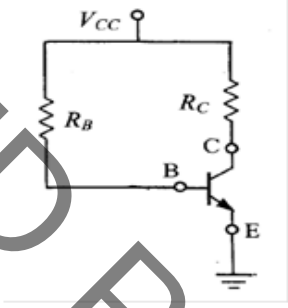
OR

7.a) Explain the leakage currents in CB and CE configurations of BJT.

b) Explain how a BJT can be used as a switch. [5+5]

8.a) Demonstrate the voltage divider biasing and calculate the stability factor for BJT.

b) For the circuit shown β = 150, VCC = 10 V, R_B = 300K Ω and R_C = 1100 Ω. [5+5]
Find:
i) Quiescent Current ICQ
ii) Quiescent Voltage VCEQ.



OR

9.a) The hybrid parameters for a transistor used in CE configuration are h_{ie} = 5kΩ; h_{fe} = 180; h_{re} = 1.25 × 10^{-4}; h_{oe} = 16 × 10^{-6} ohms. The transistor has a load resistance of 20 KΩ in the collector and is supplied from a signal source of resistance 5 KΩ. Compute the value of input impedance, output impedance, current gain and voltage gain.

b) Draw a transistor amplifier using self-bias (CE configuration) and explain the operation. [5+5]

10.a) Describe working of n-channel JFET with help of constructional diagram and draw its drain characteristic.

b) Define three FET parameter gm, rd and μ. Prove that μ= gm x rd. [5+5]

OR

11.a) Compare and contrast MOSFET with JFET.

b) With neat diagram, explain the operation of MOSFET in Depletion mode. [5+5]

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